

FUNDAMENTALS OF PLASMA PHYSICS

Fundamentals of Plasma Physics is a rigorous explanation of plasmas relevant to controlled fusion, astrophysical plasmas, solar physics, and magnetospheric plasmas, plasma thrusters, and many other plasma applications. More thorough than previous texts, it exploits new, powerful mathematical techniques to develop deeper insights into plasma behavior.

The initial chapters develop the basic plasma equations from first principles and explore single particle motion with particular attention to adiabatic invariance, a concept that later recurs in many contexts. The author then examines the many types of plasma waves and the philosophically intriguing issue of Landau damping. Magnetohydrodynamic equilibrium and stability are then tackled with emphasis on the topological concepts of magnetic helicity and self-organization. More advanced topics follow, including magnetic reconnection, nonlinear waves, and the Fokker–Planck treatment of collisions. The book concludes by discussing non-neutral and dusty plasmas, and considers how these unconventional plasmas relate to conventional plasmas.

Written for beginning graduate students and advanced undergraduates, this text emphasizes the fundamental principles that apply across many different contexts. It is of interest to students and researchers in physics, astronomy, space physics, electrical engineering, and aeronautics.

PAUL M. BELLAN is Professor of Applied Physics at the California Institute of Technology. He received his B.Sc. from the University of Manitoba and Ph.D. from Princeton University. During the last thirty years he has worked on many facets of experimental and theoretical plasma physics, including controlled fusion, industrial arcs, magnetospheric physics, solar physics, and astrophysics. He has designed and constructed several laboratory experimental devices, including a small tokamak and several spheromaks, and has worked with numerous diagnostic techniques including laser induced fluorescence, ultra-high-speed photography, and various types of probes. He has investigated Alfvén, lower hybrid, ion acoustic, and drift waves and has had a particular interest in the phenomenon of resonance cones. Recently he has built high-power pulsed laboratory devices which produce laboratory replicas of solar prominence and astrophysical jet plasmas and is using these plasmas to explore magnetohydrodynamic self-organization, magnetic helicity, and the acceleration of high speed flows.

In 2001, Professor Bellan received the Popular Writing Award from the Solar Physics Division of the American Astronomical Society for an article he wrote for *American Scientist*. His previous book, *Spheromaks*, was published in 2000. He is a Fellow of the American Physical Society and has published over 80 papers in journals including *Physical Review Letters*, *Geophysical Research Letters*, *Physics of Plasmas*, *Nuclear Fusion*, and *The Review of Scientific Instruments*.

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To my parents

